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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

Major Project

Batch: 11

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Title: AI-Driven Adaptive Transaction Batching for Cardano Decentralized Exchanges

Abstract:

Decentralized exchanges (DEXs) on the Cardano blockchain rely on off-chain batchers to aggregate user swap orders into single transactions, a requirement of Cardano's Extended UTXO (eUTXO) model, in which a shared liquidity-pool UTXO can be consumed only once per block. Existing batchers apply static rules such as fixed batch sizes or intervals and ignore real-time network conditions. Because Cardano's transaction fees are deterministic and do not rise with demand, the real cost of congestion is not higher fees but increased confirmation latency and failed transaction inclusion as blocks approach their size and execution-unit limits, which often reach 80-90% capacity during peak activity. This project develops an AI/ML-driven adaptive batching system that minimizes user waiting time and per-user cost. A forecasting model predicts near-term block fullness, while a decision policy, built as a constrained optimizer and a reinforcement-learning agent, decides when to submit a batch and how many orders to include, subject to block-capacity limits. The policy is trained and evaluated in a discrete-event simulator that replays historical Cardano blocks against modeled order arrivals, and is benchmarked against static baselines on latency, cost, inclusion success, and throughput. A minimal order-book DEX built in Aiken and deployed on the Cardano preprod testnet, with the proposed batcher as its authorized executor, provides an end-to-end demonstration.

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